Experiment 8.2

Shielding Effectiveness and EMC Compliance Analysis using MATLAB

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Aim

To calculate the shielding effectiveness of a shield material using absorption and reflection losses, and to check EMC compliance of measured EMI against the specified limit.

Software Used

MATLAB

Theory / Formulas

Absorption loss: $A = 8.686\ (t / \delta)$ dB, where t is the shield thickness and δ is the skin depth.
Reflection loss: $R = 168 - 10 \log_{10}(1 / (\mu_r f))$ dB.
Total shielding effectiveness: $SE = R + A + B$ dB, where B is the correction factor for multiple reflections.

Procedure

1. Define the shield material parameters (thickness, permeability, conductivity) and the frequency range.
2. Calculate skin depth, absorption loss and reflection loss.
3. Compute the total shielding effectiveness.
4. Compare the measured EMI with the EMC limit.
5. Plot the results and note the compliance status.

Output

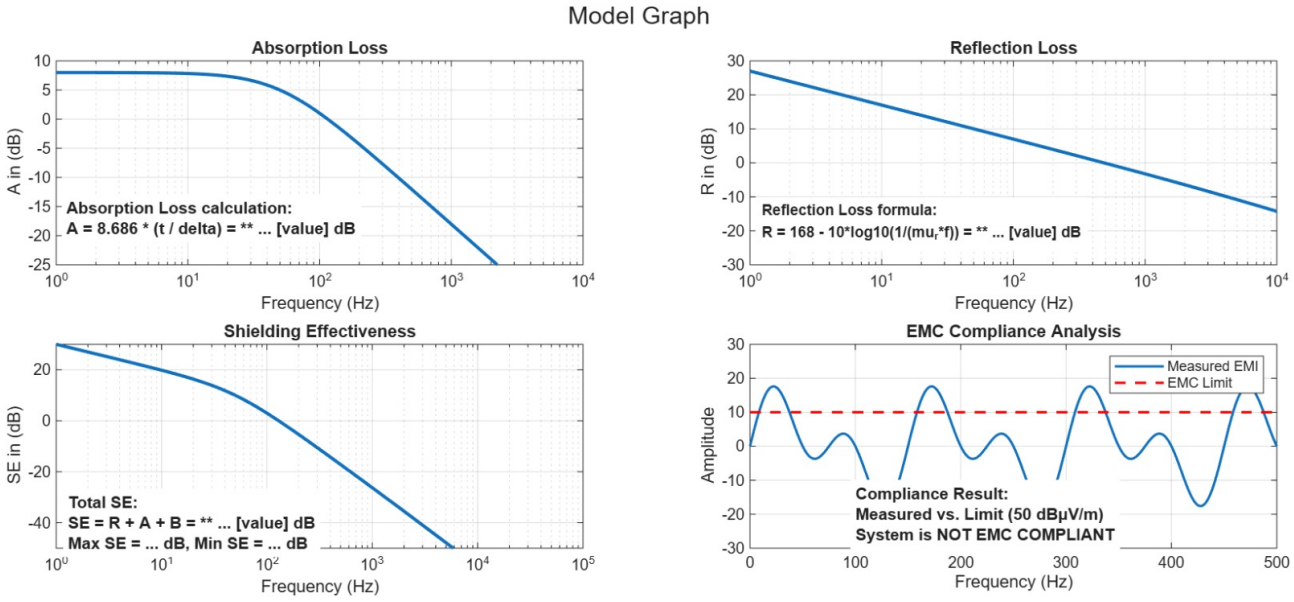


Fig. 1: Absorption loss, reflection loss, shielding effectiveness and EMC compliance analysis

Result

The absorption loss, reflection loss and total shielding effectiveness were plotted against frequency. In the EMC compliance analysis, the measured EMI exceeded the limit (50 dBμV/m), so the system is NOT EMC compliant.